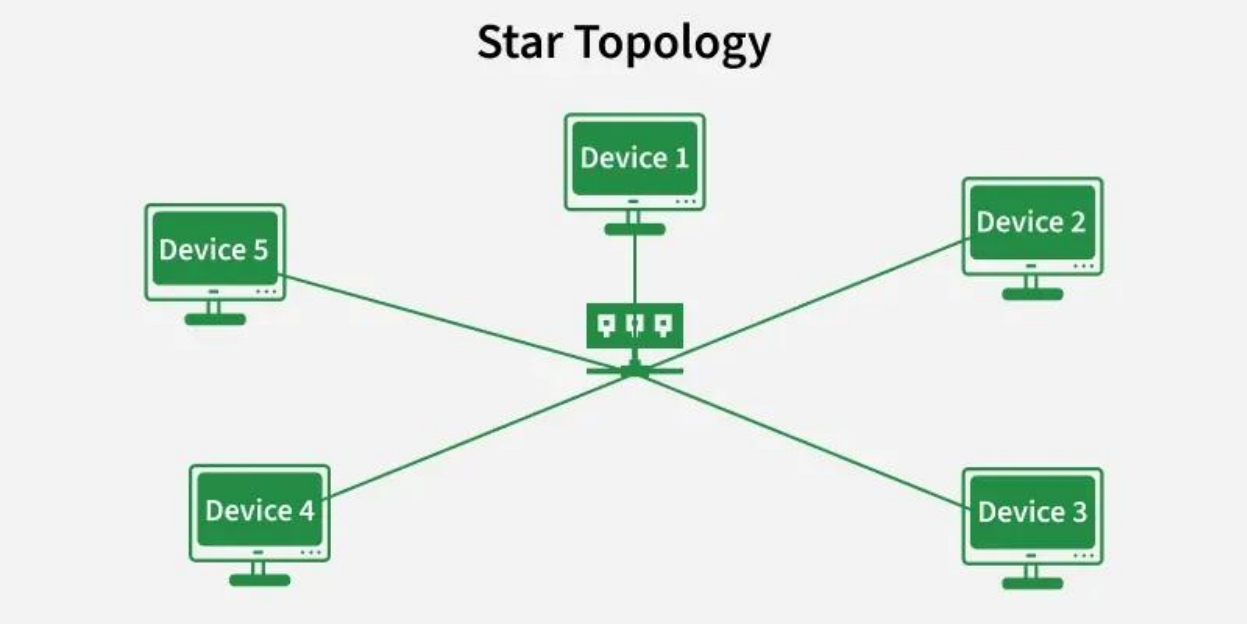
Network Topologies

Task 1 : Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

Task 1: Network Topologies

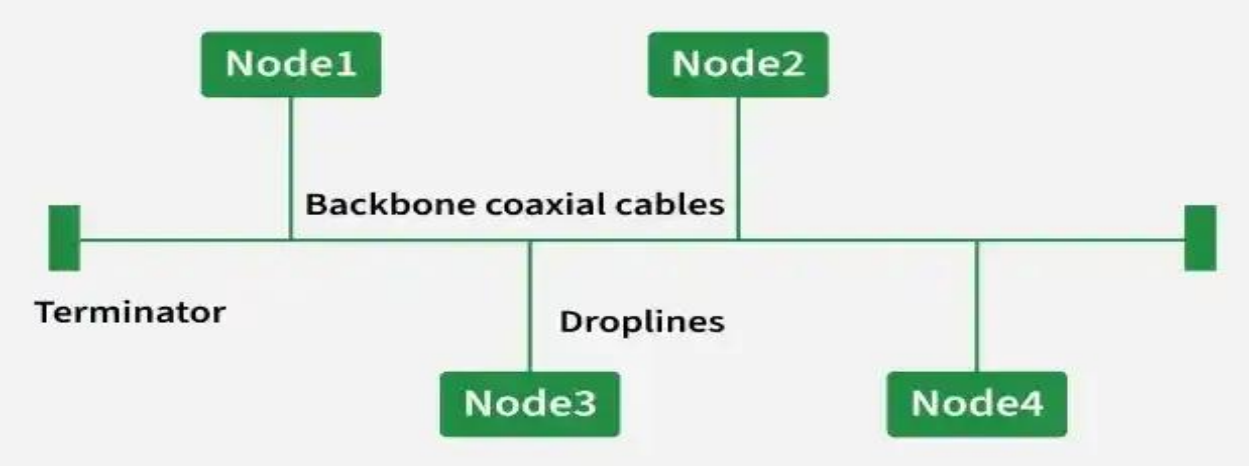
1. Star Topology



Real-World Example: Office computer network.

Explanation: In a star topology, all devices connect to a central switch. It is easy to manage and troubleshoot because a single device's connection failure usually does not affect the other devices.

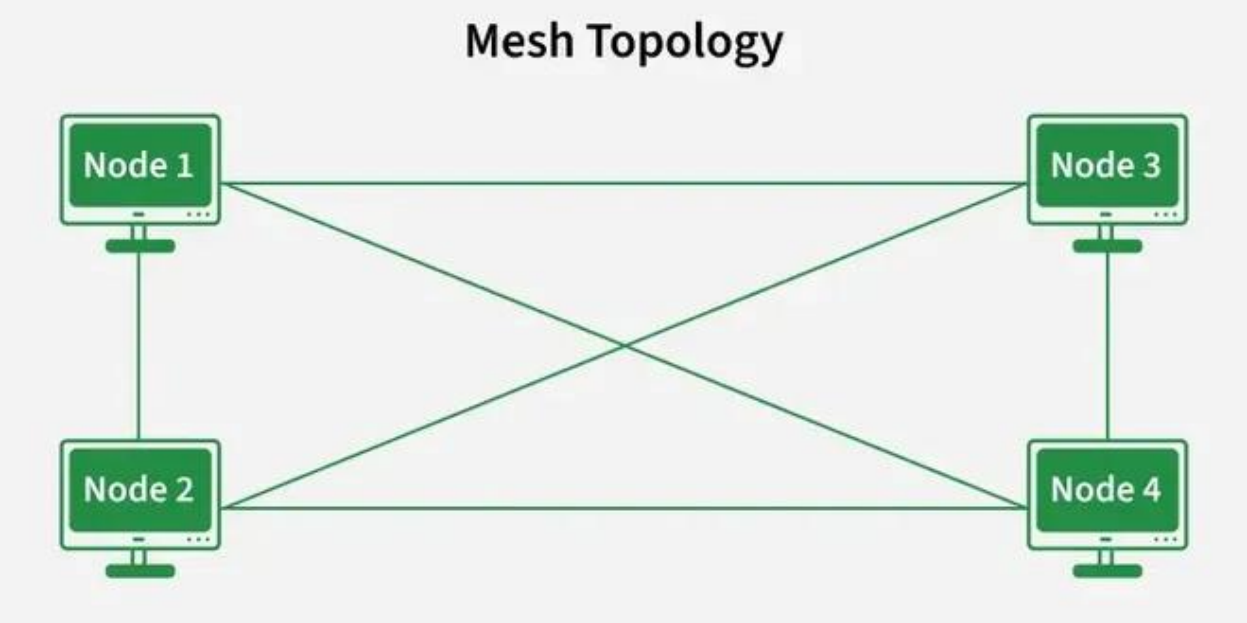
2. Bus Topology



Real-World Example: A small legacy Ethernet network.

Explanation: In a bus topology, all devices share a single backbone cable. It requires less cable, but a failure in the main cable can disrupt the entire network.

3. Mesh Topology



Each PC has a direct connection to every other PC.

Real-World Example: A wireless mesh network used to extend Wi-Fi coverage across a large building.

Explanation: Mesh topology provides multiple paths for data to travel, improving reliability. If one link fails, traffic may use another available path, although the cost and complexity are higher.

Conclusion

- Star topology is easy to manage,
- bus topology uses a shared backbone cable,
- mesh topology provides multiple communication paths for better reliability.

Task 2 : Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

1. Star Topology

Advantages (Pros):

1. Easy to install and manage.
2. Easy to identify and troubleshoot network problems.
3. Failure of one device or cable usually does not affect other devices.
4. Adding or removing devices is simple.

Disadvantages (Cons):

1. Requires more cables than bus topology.
2. Needs a central switch or hub.
3. Failure of the central switch can stop the entire network.
4. Installation costs can be higher because of the central device and cabling.

Best Use Case: Offices, schools, and home LANs.

App/Service Example: Instagram used in an office network. Office computers can connect through a star-topology LAN to access Instagram and other internet services.

2. Bus Topology

Advantages (Pros):

1. Requires less cable than many other wired topologies.
2. Simple to set up for a small network.
3. Does not require a central switch.
4. Can be inexpensive for small, basic networks.

Disadvantages (Cons):

1. Failure of the backbone cable can disrupt the whole network.
2. Performance decreases as traffic and the number of devices increase.
3. Troubleshooting cable faults can be difficult.

4. Adding more devices can disrupt the network and requires careful cable termination.

Best Use Case: Small legacy networks and basic setups where low cabling cost is important.

App/Service Example: A legacy Ethernet LAN used by computers to access a shared file server.

3. Ring Topology

Advantages (Pros):

1. Each device has a predictable path for data transmission.
2. Can provide orderly access to the network in token-based systems.
3. May avoid collisions in token-passing implementations.
4. Can provide predictable performance under controlled traffic conditions.

Disadvantages (Cons):

1. A single link or device failure can disrupt a basic ring.
2. Adding or removing devices may interrupt the network.
3. Troubleshooting can be complicated.
4. It is less flexible and less common in modern general-purpose LANs.

Best Use Case: Specialized industrial or telecommunications networks designed around ring architectures.

App/Service Example: A ring-based industrial network carrying monitoring data between connected control devices.

4. Mesh Topology

Advantages (Pros):

1. Provides multiple paths between devices.
2. Offers high reliability and fault tolerance.
3. A failed link does not necessarily stop communication.
4. Supports alternative routes when a connection becomes unavailable.

Disadvantages (Cons):

1. Requires more cabling or wireless links.
2. Installation and maintenance can be expensive.
3. Configuration and troubleshooting can be complex.
4. A full mesh becomes difficult to scale because the number of links grows rapidly.

Best Use Case: Wireless mesh networks, critical infrastructure, and networks requiring high availability.

App/Service Example: A wireless mesh network that provides Wi-Fi coverage across a large campus or building.

5. Comparison Table

Topology	Main Advantage	Main Disadvantage	Best Use Case	Example
Star	Easy management and troubleshooting	Central switch is a single point of failure	Offices and home LANs	Office computers accessing Instagram
Bus	Low cabling requirements	Backbone failure can disrupt the network	Legacy small LANs	Older shared-cable Ethernet
Ring	Predictable communication in token-based systems	A basic ring can fail if a link breaks	Specialized industrial networks	Ring-based industrial monitoring
Mesh	High reliability and alternative paths	High cost and complexity	Large or critical networks	Campus wireless mesh

Conclusion

Each network topology has different advantages and disadvantages. Star topology is suitable for most modern LANs, bus topology is associated mainly with legacy networks, ring topology is useful in specialized systems, and mesh topology is preferred when reliability and alternative communication paths are important.

Task 3 : Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

Game Selected: PUBG Mobile

Chosen Topology: Mesh Topology between game servers and network infrastructure.

Justification:

- I would choose a mesh-based network architecture between important game servers and network infrastructure because it provides multiple communication paths and improves reliability.
- If one network link fails, traffic can be redirected through another available path, helping maintain stable gameplay.
- Proper routing and nearby game servers can reduce latency and improve response speed.
- However, a full mesh can be expensive and complex, so a partial mesh is more practical and cost-effective for a large multiplayer game.

Conclusion: A partial mesh network is a suitable choice for supporting reliable communication and minimizing disruptions in a multiplayer game like PUBG Mobile.

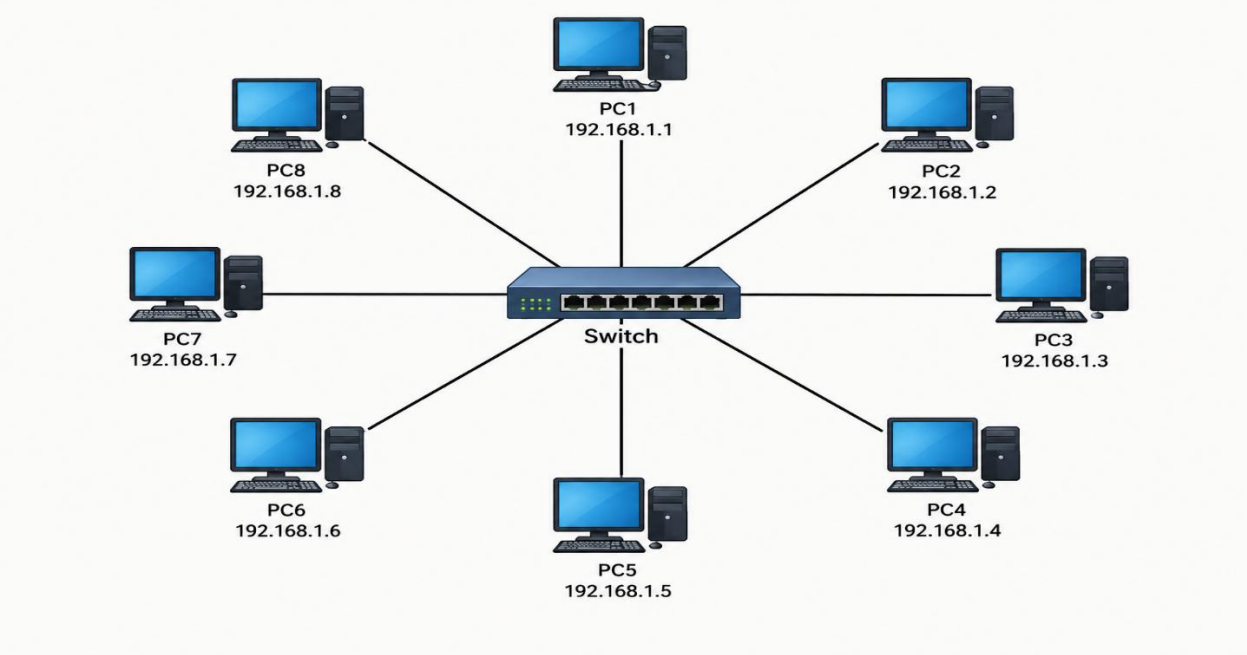
Task 4 : Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.

Scenario

A small warehouse wants to connect 8 computers for order processing and inventory updates.

Recommended Topology: Star Topology



Note: All eight computers are connected directly to a central network switch using Ethernet cables.

Explanation

I recommend star topology because it is cost-effective, easy to install, and simple to troubleshoot. If one computer or its cable fails, the other computers can continue working normally. It is also easy to expand the network by connecting additional computers to the switch, provided enough ports are available.

Conclusion

Star topology is the most suitable choice for this warehouse because it offers a good balance of cost, reliability, easy maintenance, and future expansion.